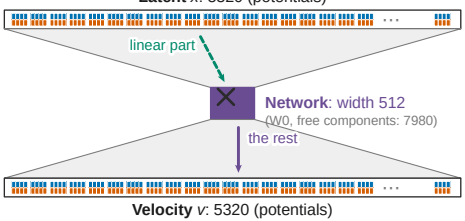
a **Optimal velocity field $v(x, t)$**

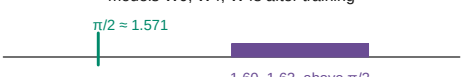
v = **Linear part** $A(t)\,x + b(t)$ + **The rest** non-linear

Gaussian data: exactly affine — Gaussian data: none



Conjecture (tested by a width scan, H21): a network narrower than the latent cannot carry the linear part

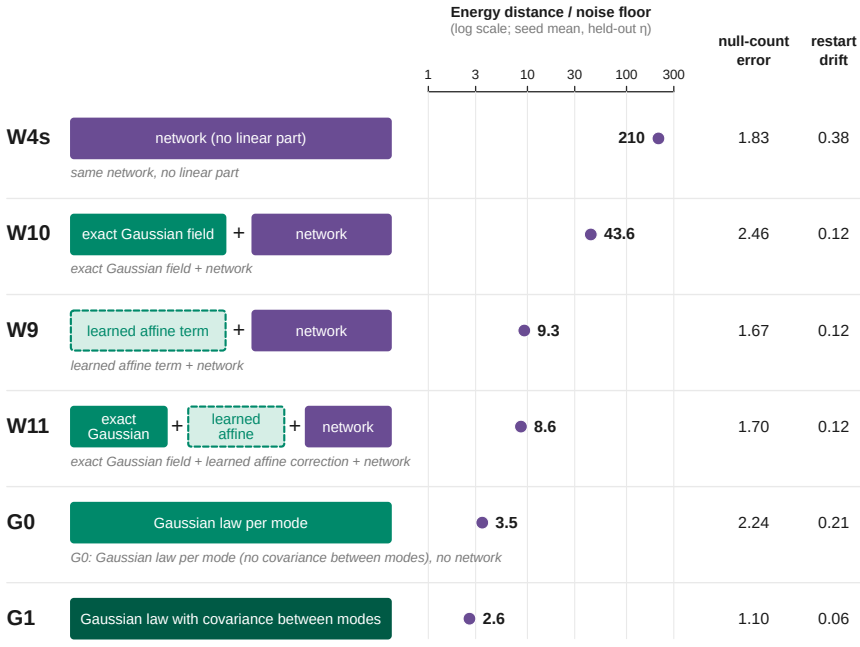
Measured: flow-matching loss of the plain models W0, W4, W4s after training



Samples over-dispersed: potentials models: 2 to 3 times too many null points

b **Four remedies and two Gaussian baselines**

(relative values)



Two Gaussian laws without a network are closer to the reference than every learned model; the one that keeps the covariance between modes (G1) is the best on all three columns.